

HOMEWORK 2
due Wed, Sep. 16, 11:59 PM
(Upload your solutions to Gradescope)

1. (3 points) Let A be an $m \times n$ matrix. Prove that

$$\|A\|_{\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|.$$

2. (2 points) Suppose $A : V \rightarrow W$ and $B : U \rightarrow V$ are bounded operators. Show that $(AB)^* = B^*A^*$. If A is invertible, show that A^* is invertible and $(A^*)^{-1} = (A^{-1})^*$.
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3. (3 points) (#12 p. 35. Stop at x^2 , i.e., drop x^3 from the list.) By using the Gram-Schmidt orthogonalization procedure, find an orthonormal basis for the subspace of $L^2[0, 1]$ spanned by $1, x, x^2$.
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4. (3 points) (#13 p. 36. Stop at x^2 and replace $\cos x$ by $\cos(\pi x)$.) Find the $L^2[0, 1]$ projection of the function $\cos(\pi x)$ onto the space spanned by $1, x, x^2$.
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5. (3 points) (#15 p. 36.) Project the function $f(x) = x$ onto the space spanned by $\phi(x), \psi(x), \psi(2x), \psi(2x - 1) \in L^2[0, 1]$, where

$$\phi(x) = \begin{cases} 1, & 0 \leq x < 1, \\ 0, & \text{otherwise,} \end{cases} \quad \psi(x) = \begin{cases} 1, & 0 \leq x < 1/2, \\ -1, & 1/2 \leq x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

6. (2 points) (#19 p. 36.) Suppose $K(x, y)$ is a continuous function that vanishes outside a bounded set in $\mathbb{R}^2$. Define $T : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ by

$$Tf(x) = \int_{-\infty}^{\infty} K(x, y)f(y) dy.$$

Show that $T^*g(x) = \int_{-\infty}^{\infty} \overline{K(y, x)}g(y) dy$. Note the parallel with the adjoint of a matrix ($A_{ij}^* = \overline{A_{ji}}$).

7. (2 points) (#20 p. 36.) Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Show that $\text{Ker}(A^*) = (\text{Range}(A))^\perp$. Note: Ker stands for Kernel; $\text{Ker}(A^*)$ is the set of all vectors in W which are sent to zero by A^* .

8. (2 points) (#21 p. 37. Assume V is finite dimensional—the theorem is not generally true otherwise.) Prove the following theorem (Fredholm's alternative): Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Let b be any element in W . Then either

- $Ax = b$ has a solution $x \in V$; or
 - There is a vector $w \in W$ with $A^*w = 0$ and $\langle b, w \rangle_W \neq 0$.
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Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggress and Narcowich.